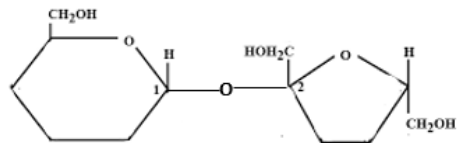


Question number	Answer	Additional guidance	Mark
3(a)(i)	O joined to C1 on glucose and C2 on fructose by covalent bond (1)	 ACCEPT bonds not touching carbons provided it is clear where they are supposed to be attached to solid or dotted lines for covalent bond DO NOT ACCEPT other groups added to C1 or C2 IGNORE water shown other groups added to rest of molecule	(1)

Question number	Answer	Additional guidance	Mark
3(a)(ii)	An answer that makes reference to the following points: $C_{12}H_{22}O_{11}$ (1) H_2O (1)	ACCEPT numbers that are not subscript numbers IGNORE if not written as part of the equation	(2)

Question number	Answer	Additional guidance	Mark
3(b)(i)	sucrose contains {the same number of / one } fructose (1)	ACCEPT ratio of sucrose to fructose is 1 : 1 as the concentration of sucrose increases so does the (concentration of) fructose the more sucrose there is the more fructose there will be more fructose for the chemical to bind to the same amount of chemical binds to each fructose DO NOT ACCEPT glucose	(1)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	the test is not sensitive enough / the colour intensity is too low (1)	ACCEPT a certain concentration of {fructose / sucrose} is necessary for the {colour to be visible / colour change} equipment is not sensitive enough colour change visible above concentrations $50 \mu\text{g dm}^{-3}$ sucrose DO NOT ACCEPT glucose	(1)

Question number	Answer	Additional guidance	Mark
3(b)(iii)	the chemical (only) binds to fructose / there is no fructose in either maltose or lactose (1)	ACCEPT chemical cannot bind to maltose or lactose	(1)